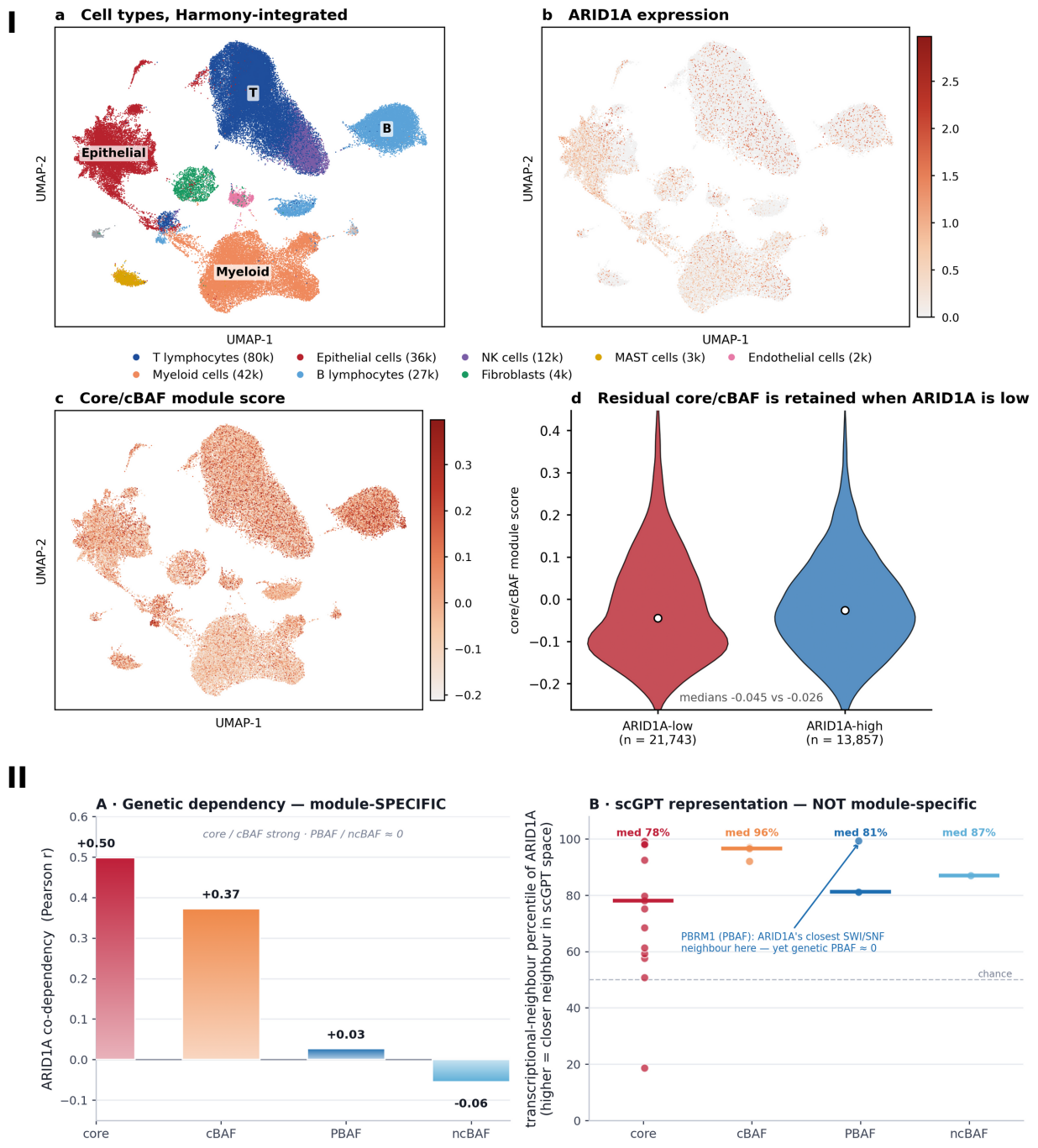

I
a Cell types, Harmony-integrated
T
B
Epithelial
Myeloid
UMAP-1
UMAP-2
b ARID1A expression
2.5
2.0
1.5
1.0
0.5
0.0
T lymphocytes (80k)
Epithelial cells (36k)
NK cells (12k)
MAST cells (3k)
Endothelial cells (2k)
Myeloid cells (42k)
B lymphocytes (27k)
Fibroblasts (4k)
c Core/cBAF module score
0.3
0.2
0.1
0.0
−0.1
−0.2
d Residual core/cBAF is retained when ARID1A is low
core/cBAF module score
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
medians -0.045 vs -0.026
ARID1A-low
(n = 21,743)
ARID1A-high
(n = 13,857)
II
A · Genetic dependency — module-SPECIFIC
core / cBAF strong · PBAF / ncBAF ≈ 0
ARID1A co-dependency (Pearson r)
0.6
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
+0.50
+0.37
+0.03
-0.06
core
cBAF
PBAF
ncBAF
B · scGPT representation — NOT module-specific
transcriptional-neighbour percentile of ARID1A
(higher = closer neighbour in scGPT space)
med 78%
med 96%
med 81%
med 87%
PBRM1 (PBAF): ARID1A's closest SWI/SNF
neighbour here — yet genetic PBAF ≈ 0
chance
100
80
60
40
20
0
core
cBAF
PBAF
ncBAF